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Playback speed and learning: effects on objective comprehension and perceived clarity

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Introduction: Playback Speed (PS)—the practice of reproducing audiovisual content at speeds higher than standard—has established itself as a common strategy in virtual learning environments, though its effects on the learning experience are not fully understood. The empirical literature shows polarized results, and prior research has focused almost exclusively on objective performance while ignoring the subjective processing experience, as well as overlooking sociodemographic moderators such as age and technological familiarity. This study examined the effects of Playback Speed on objective information comprehension and perceived information clarity, further evaluating the moderating role of age and technological familiarity.

Methods: A quantitative between-subjects experimental design with one factor was applied, using three speed conditions (1.0x, 1.5x, and 2.0x). A sample of 328 participants, distributed across three generational cohorts (18–73 years), was exposed to a standardized pre-recorded technical lesson on digital marketing of moderate duration (16 min 42 s at 1.0x). Objective comprehension was assessed through a 10-item multiple-choice test, while perceived clarity and technological familiarity were measured using validated Likert-scale instruments. Data were analyzed using Partial Least Squares Structural Equation Modeling (PLS-SEM) with 5,000 bootstrapping subsamples.

Results: Playback Speed did not significantly impair objective information comprehension in any of the evaluated conditions ($R^2 = 0.022$). However, it produced a progressive and statistically significant decline in perceived clarity, particularly at 2.0x ($\beta = -0.311$, $p < 0.001$; $f^2 = 0.104$), while the effect at 1.5x was marginal and lacked practical significance. Neither technological familiarity nor age moderated the effect of Playback Speed on comprehension. The only significant interaction found (age $\times$ 1.5x on perceived clarity) was of small magnitude ($f^2 = 0.013$) and should be interpreted with caution.

Discussion: These findings highlight a dissociation between objective learning outcomes and the subjective processing experience: Playback Speed operates as an asymmetric exchange that preserves retention but increases the perceived cost of processing. This pattern is consistent with Cognitive Load

Theory, suggesting that moderate accelerations may not exceed the working memory overload threshold required to disrupt encoding, yet are sufficient to degrade processing fluency. The results have implications for the design of digital educational content and for guiding students who use acceleration as a time management strategy.

KEYWORDS

information clarity, information comprehension, playback speed, speedwatching, virtual education

1 Introduction

Currently, society is immersed in a volume of digital content that exceeds human consumption capacity, and as a direct response to this pressure, students and content consumers adopt strategies to optimize their learning processes (Imundo et al., 2025). Simultaneously, technological platforms—from streaming giants like Netflix to online learning environments (MOOCs) like Coursera—have responded to this demand by integrating features that grant users control over their viewing experience (Kim et al., 2025). Within this set of tools, the function to control Playback Speed has become omnipresent, giving rise to a behavioral phenomenon known as Playback Speed (PS) or speedwatching (Keehr and Reardon, 2025). This practice, defined as the consumption of audiovisual content at speeds higher than normal, has become popularized as a behavioral adaptation to the abundance of available content (Merhavy et al., 2023). Technically, Playback Speed allows viewers to reproduce lectures, news, or educational material at speeds that commonly range between 1.25x and 2.0x, although some users report consuming content at 2.5x or higher (Chen et al., 2024).

The growing adoption of this practice has sparked interest within the scientific community, giving rise to a field of study oriented toward understanding its implications (Cheng et al., 2022). The premise driving this phenomenon is the pursuit of efficiency, understood as the ability to process more information in less time (Ahn and Chan, 2025). User motivations are diverse, ranging from practical benefits such as time-saving to perceptions of cognitive advantages, such as a purported improvement in learning and attention (Tharumalingam and Risko, 2025). However, the empirical literature on the effects of Playback Speed yields highly polarized results. On one hand, some studies suggest that speeds up to 2.5x do not affect objective retention and may even optimize learning at moderate speeds (1.25x–1.5x) by reducing mind-wandering (Ahn and Chan, 2025; Kiyak et al., 2023). On the other hand, recent research and experimental analyses warn of significant impairments in performance and memory starting at thresholds of 1.5x and 2.0x (Liu and Jia, 2025; Tharumalingam et al., 2025; Tran et al., 2024).

This polarization highlights that the impact of acceleration is not universal, revealing critical research gaps that prevent a comprehensive understanding of the phenomenon. In the first place, studies have focused almost exclusively on objective performance (comprehension), ignoring “perceived clarity.” Distinguishing between both constructs is fundamental and warrants separate investigation: while comprehension measures pure data retention, perceived clarity captures the metacognitive

experience and the user’s fluency. Understanding this latter dimension is vital because a confusing or arduous experience, even if it results in learning, can generate a false sense of failure that undermines student self-efficacy and long-term motivation (Keehr and Reardon, 2025). Secondly, in the current context of virtual education and lifelong learning, it is problematic to assume that all users process accelerated information equally. Sample homogeneity is a major limitation, as research has centered on cohorts of young university adults (Murphy et al., 2023). This leaves a profound gap regarding how age, which naturally slows neurological processing speed, alters cognitive resilience when faced with accelerated educational video. Closely linked to age, a third gap emerges: technological familiarity. Until now, research has ignored how a user’s prior digital competence moderates the cognitive load of interacting with platforms at high speeds, erroneously assuming that all users possess the same proficiency (Kuhlmann et al., 2024).

In synthesis, although Playback Speed has become a standard, it remains unknown whether the objective assimilation of data dissociates from the subjective perceptual experience, and under which sociodemographic conditions this practice ceases to be efficient and becomes counterproductive. Resolving these unknowns is imperative for educators and instructional designers to create equitable and effective online learning environments. Therefore, the present study specifically tests how Playback Speed simultaneously affects objective information comprehension and perceived clarity in the context of a virtual technical class, introducing age (18 to 73 years) and technological familiarity as moderating variables. By rigorously contrasting learning with user experience and controlling for digital competencies, this study seeks to untangle the contradictions in the literature and offer a precise model of who truly benefits from content acceleration.

2 Theoretical framework

2.1 Playback speed: definition and cognitive adaptation in the digital era

PS is defined as the practice of reproducing audiovisual content at a speed higher than normal, usually between 1.5 and 2 times the standard playback speed (Wilson et al., 2018). This technique has emerged as a response to the overwhelming amount of information available in the digital age and the need to process this information more efficiently. Originating within the technological community, PS has been rapidly adopted by a

broader audience, becoming a predominant trend, especially among young people (Lang et al., 2020).

Historically, media consumption occurred at a pace pre-established by content creators; however, with the advent of digital platforms and streaming services, viewers have gained greater control over the way they consume information (De'Sperati and Baud Bovy, 2017). YouTube, Netflix, and other platforms have incorporated options that allow users to adjust the playback speed, facilitating PS. This practice is not limited solely to entertainment but has also extended to the educational field and news consumption, where users seek to optimize their time and absorb large volumes of information rapidly (Cheng et al., 2021).

The popularity of PS reflects a broader trend toward efficiency and time optimization in modern life. Young people, in particular, resort to this technique to stay up-to-date in a world where information is constantly updated and the pressure to stay informed is high (Wingralek et al., 2022), mitigating contemporary psychological pressures such as FOMO (Fear Of Missing Out). Nevertheless, this phenomenon also raises important questions about its long-term effects on comprehension and information retention, as well as on the general user experience. The practice of PS might be changing the way we process information, challenging our cognitive capacities and altering our perception of viewed content (Murphy et al., 2023).

2.2 Cognitive load theory and limited processing: links to comprehension and perceived clarity

Research on the impact of PS on learning outcomes is characterized by a marked contradiction (Murphy et al., 2023). On one hand, a current of studies suggests that acceleration has minimal or no effects on comprehension. Some findings show that speeds up to 2.0x or 2.5x do not significantly affect information retention (Chen et al., 2024; Kıyak et al., 2023), and an optimization of learning has even been reported at moderate speeds (1.25x or 1.5x), possibly because a faster pace demands greater attention and reduces mind-wandering (Ahn and Chan, 2025; Murphy et al., 2023). On the other hand, an equally solid line of research warns about the detriments of acceleration. A recent meta-analysis concluded that increasing playback speed negatively affects exam performance (Tharumalingam et al., 2025), and some experimental studies support this conclusion, pointing to performance decrements even at speeds of 1.5x (Liu and Jia, 2025) and memory impairment when exceeding the 2x threshold (Tran et al., 2024).

This apparent paradox can be explained through Cognitive Load Theory (CLT) proposed by Sweller (1988) and limited information processing models (Kintsch, 1988). Rather than observing learning as an “all or nothing” phenomenon, CLT allows us to untangle how mental effort is divided between pure retention and the processing experience (Wilson et al., 2018). This theory maintains that human working memory is limited. As PS increases, both intrinsic load (more units of information per second) and extrinsic load (the additional effort of decoding compressed speech and visual elements) increase (Mo et al., 2022). It is likely that an optimal threshold exists: moderate speeds could optimize cognitive load by keeping the user focused

and reducing mind-wandering, but exceeding the individual threshold produces an overload that saturates working memory and prevents the proper encoding of information into long-term memory, thus affecting comprehension (Kintsch, 1988). In this sense, limited information processing postulates that the cognitive system requires physical and mental time to perform essential operations such as speech segmentation. When extrinsic overload exhausts valuable resources intended for the construction of knowledge schemas, it inevitably results in a decrease in encoding (Cárdenas, 2018). Therefore, it is proposed that:

H1: Playback Speed has a negative effect on objective information comprehension when it exceeds certain thresholds.

Beyond objective information comprehension, this study analyzes perceived information clarity, understood as the user's subjective evaluation of how easy to follow, organized, and understandable the presented information is (Sidelinger and McCroskey, 1997). In this study, perceived clarity is conceptualized as a dual construct: on one hand, it has a communicational nature reflecting the structure of the message; on the other hand, it is a deeply metacognitive construct representing the individual's subjective self-assessment of their own processing fluency (Keehr and Reardon, 2025). While comprehension evaluates the processing outcome, clarity reflects the subjective experience of it. As PS increases, the cognitive effort required to decode the message also intensifies. Although the user may maintain comprehension through greater attentional effort, processing fluency decreases. Divided attention and the constant need to reprocess accelerated stimuli interrupt the natural flow of cognition (Pérez et al., 2024). This perceived difficulty can reduce the sense of control and confidence in what has been learned and lead to a negative evaluation of message clarity (Keehr and Reardon, 2025). Consequently, the following hypothesis is proposed:

H2: Playback Speed has a negative effect on the perceived information clarity when it exceeds certain thresholds.

2.3 Moderation logic: effects of technological familiarity and age

The contradictions in the literature suggest that the relationship between PS and comprehension is not universal but is instead mediated by individual variables (Liu and Jia, 2025; Mo et al., 2022). This study focuses on two key moderators that have been underestimated: age and technological familiarity. Linked to age, technological familiarity (TF) emerges as a potential confounding factor. TF is conceptualized here within the framework of Computer Self-Efficacy Theory (Compeau and Higgins, 1995) and digital literacy, where a high level of familiarity implies that the individual has developed automated schemas to process information in digital environments (Getenet, 2024). The difficulties attributed to cognitive aging might be exacerbated by lower fluency in navigating digital interfaces. Applying Cognitive Load Theory, interacting with an unfamiliar tool (such as speed controls) imposes an additional extrinsic cognitive load (Sweller,

1988). This extraneous load consumes working memory resources that would otherwise be available for processing the lesson content. Conversely, an individual with high TF manages the interface automatically and without anxiety, dedicating their cognitive resources fully to learning and mitigating the negative impact of PS. Therefore, low TF would aggravate the negative effects of PS, while high TF could mitigate them by allowing for effective comprehension at higher speeds. Based on this, the following hypothesis is proposed:

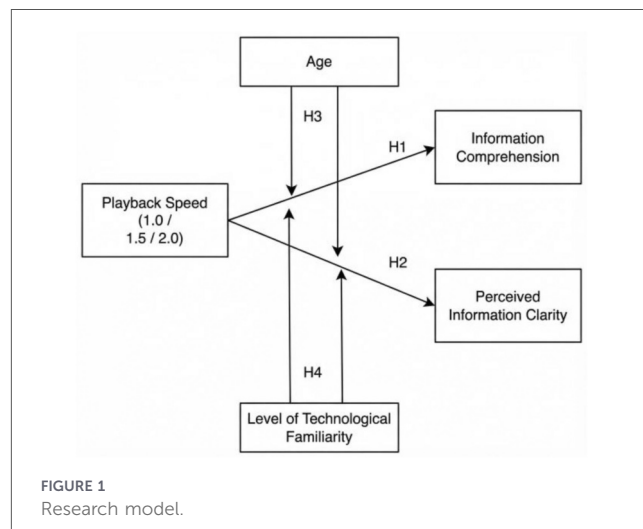
H3: Technological familiarity moderates the effect of Playback Speed on objective information comprehension and perceived clarity such that the negative effect will be more pronounced in individuals with low technological familiarity.

On the other hand, age is a critical moderator given that most studies have focused almost exclusively on young adults (Kiyak et al., 2023). Processing-Speed Theory postulates a general slowing of basic cognitive and neurological operations with aging (Salthouse, 1996). This slowing implies that, since older adults intrinsically require more time to process sensory-perceptual tasks, the cognitive overload “threshold” is reached at lower playback speeds in older adults (Park and Reuter-Lorenz, 2009; Sweller, 1988). Evidence supports this: Murphy et al. (2023) found that while young adults showed no memory deficits with PS, the performance of older adults was indeed affected. Similarly, Sheng et al. (2022) observed age-differentiated effects. Therefore, the following hypothesis is proposed:

H4: Age moderates the effect of Playback Speed on objective information comprehension and perceived clarity such that the negative effect will be more pronounced in older adults.

2.4 Conceptual synthesis of the research model

Overall, the presented theoretical framework articulates a system of interdependent relationships that seeks to explain the phenomenon of PS. As illustrated in Figure 1, the model postulates that reproduction speed acts as the central cognitive stressor (independent variable). The effects of this acceleration bifurcate into two learning dimensions that respond to different processing logics: objective information comprehension (as a measurable cognitive outcome, H1) and perceived clarity (as a metacognitive and communicational experience, H2), which constitute the dependent variables. In turn, this system does not operate in a vacuum but is conditioned by the individual characteristics of the user: technological familiarity (H3) and age (H4) function as moderating variables. These variables intervene orthogonally upon the main effects, theoretically suggesting that they possess the capacity to buffer (in the case of high familiarity) or exacerbate (in the case of older age) the impact of the cognitive stressor on learning and perception. The direct arrows in Figure 1 represent the expected direct effects, while the perpendicular arrows mathematically illustrate the



interaction effects proposed in the moderation hypotheses. The proposed research model is presented below in Figure 1.

3 Materials and methods

3.1 Research design and ethical considerations

To evaluate the effects of PS on objective information comprehension and perceived information clarity, as well as to explore the moderating role of age and technological familiarity, a quantitative between-subjects experimental design with one factor (Playback Speed: 1.0x, 1.5x, and 2.0x) was applied. This design is appropriate because it combines the active manipulation of the independent variable with the random assignment of participants to experimental conditions, allowing for the establishment of causal relationships and the systematic control of extraneous variables (Gerber and Green, 2012; Shadish, Cook and Campbell, 2002). Although the sample was recruited through an online convenience panel, randomization operated at the level of individual assignment to groups—a criterion that, following Campbell and Stanley (1963), is the defining element of a true experiment. To ensure ethical rigor, the research adhered to institutional guidelines and the Declaration of Helsinki. Before starting, all participants read and accepted a digital informed consent form on the platform, which guaranteed anonymity, strict data confidentiality, and the voluntary nature of their participation.

3.2 Participants and sampling

The total sample consisted of 328 participants ($n = 328$) distributed nationwide in Spain. During the period between October 5 and 15, 2025, participants were assigned in a balanced manner to three experimental conditions: Group 1 (1.0x): 110 people (33.54%); Group 2 (1.5x): 106 people (32.32%); and Group 3 (2.0x): 112 people (34.15%). Given that the impact of aging on cognitive processing is a central axis of the study, the

characteristics of the sample were directly aligned with this purpose, achieving an equitable representation of three generational cohorts that formed part of the conceptual framework: young adults aged 18 to 30 ($n = 104$; 31.7%), middle-aged adults aged 31 to 45 ($n = 116$; 35.4%), and older adults aged 46 to 73 ($n = 108$; 32.9%), with an overall mean age of 39.14 years. Recruitment and fieldwork were outsourced to a specialized market research firm, which administered an online panel under a non-probability convenience sampling scheme. Its technical infrastructure guaranteed the automated random assignment of participants to the different experimental conditions. To ensure the internal validity of the study, baseline demographic equivalence was verified. A one-way analysis of variance (ANOVA) confirmed that there were no statistically significant differences in the average age between the groups assigned to 1.0x, 1.5x, and 2.0x [$F(2, 325) = 0.84$, $p = 0.433$]. As detailed in Table 1, this empirically demonstrates that the three playback conditions were completely comparable from the baseline.

It is also important to note that to control for potential confounding variables, an initial filter was implemented to exclude any individual with prior knowledge or academic background in the video's topic, thus ensuring a cognitive "blank slate" to avoid bias in the comprehension results.

3.3 Materials and procedure

Participants were exposed to a standardized pre-recorded class on digital marketing, specifically focused on the construction and implementation of e-commerce websites. This topic was selected for its contemporary relevance and its technical nature, which allows for the precise evaluation of conceptual comprehension. The content covered: the essential infrastructure for website operation and the role of hosting platforms; the risks of not having a private domain (loss of brand credibility, SEO limitations, and dependence on free platforms); hosting services in their various modalities (shared, VPS, and dedicated), with an emphasis on availability, security, and scalability; content management systems (CMS) such as WordPress or PrestaShop, highlighting their ease of use, plugin ecosystems, and development agility; and payment gateways as an indispensable component for validating payment methods, applying robust authentication, and mitigating fraud risks. The video was developed using everyday language and followed a progressive educational flow, from the creation of the digital environment to the full operation of the virtual store. The standard duration of the video (1.0x) was 16 min 42 s. Since the intrinsic objective of PS is efficiency, the exposure time inherently varied

according to the experimental condition: the 1.5x group completed the video in approximately 11 min and the 2.0x group in approximately 8 min. This difference in exposure time does not constitute a confounding variable, but rather the central operational mechanism of the experimental manipulation. To ensure fairness in the evaluation, all groups had unlimited time to complete the post-test questionnaire.

3.4 Instruments and validation

Prior to the fieldwork, a pilot pretest was conducted with 25 people to validate the translation and wording of the items and to confirm that both the scales and the video were understandable for the wide age spectrum of the sample. The instruments used are described below: a) objective information comprehension (dependent variable): This was evaluated through a 10-question multiple-choice test with five alternatives each, designed and validated by five professors with doctorates and scientific publications in marketing. The questions were extracted directly and exclusively from the video script; thus, the pilot pretest confirmed they measured objective retention of the content and could not be answered through intuition or general knowledge. Each item was weighted at 0.5 points, with a maximum score of 5 points. b) perceived information clarity (dependent variable): This was measured as a construct using an adaptation of the Sidelinger and McCroskey (1997) scale, which is widely validated in communication literature. The scale consisted of 5 items evaluated on a 5-point Likert scale (1 = Strongly disagree, 5 = Strongly agree). c) Technological familiarity (moderating variable): This was measured through an adaptation of the Kirsch et al. (1998) instrument, composed of 4 items that evaluate self-efficacy and the frequency of use of technological tools. Responses were coded on a 5-point Likert scale adjusted to the nature of each item (from 1 = Very poor/Very uncomfortable/Never, to 5 = Very good/Very comfortable/Daily). d) Age (moderating variable): This was collected as a continuous sociodemographic variable. The extraction of scales b) and c) from academic literature ensures their initial construct validity, while the pilot pretest allowed for verification of their suitability for the specific context of this research.

4 Results

4.1 Sample description

The sample was selected using a non-probabilistic approach, collecting a total of 328 participants ($n = 328$), a size considered sufficient to ensure the validity of the quasi-experimental analysis. The sample exhibited a diverse demographic distribution. Regarding gender, participation was nearly equitable, consisting of 50.30% men ($n = 165$), 49.39% women ($n = 162$), and 0.30% ($n = 1$) identifying with another gender.

The educational level of the participants varied, with a majority holding a professional degree (43.29%, $n = 142$). Secondary/primary education levels (22.26%, $n = 73$) and technical/technological education (21.65%, $n = 71$) had similar representation, while 12.80% ($n = 42$) had completed

TABLE 1 Analysis of variance (ANOVA) for Age by experimental group.

Source of variation	Sum of squares	df	Mean square	F	p
Between groups	314.52	2	157.26	0.84	0.433
Within groups	60,842.15	325	187.20		
Total	61,156.67	327			

df = degrees of freedom. A p -value > 0.05 confirms demographic equivalence regarding age across the experimental groups.

postgraduate studies. In terms of socioeconomic status, the sample was primarily concentrated in levels 3 (40.55%, $n = 133$) and 4 (28.35%, $n = 93$), which together accounted for 68.9% of the total.

For the moderating variable of age, it was analyzed as a continuous variable. The age range of participants was 18 to 73 years, with an average age of 39.14 years and a standard deviation of 12.21. Finally, participants were evenly assigned to the three experimental conditions (video speed). Group 1 (1.0x speed) included 110 people (33.54%), Group 2 (1.5x speed) had 106 participants (32.32%), and Group 3 (2.0x speed) consisted of 112 participants (34.15%) (See Table 2).

4.2 Data analysis

To test the hypotheses, Partial Least Squares Structural Equation Modeling (PLS-SEM) was utilized through SmartPLS 4 software. Although the between-subjects experimental design would conventionally suggest the use of ANOVA or general linear models, the choice of PLS-SEM is justified by three reasons. First, it allows for the simultaneous evaluation of the measurement model—validating the reliability and validity of reflective latent constructs (perceived clarity and technological familiarity)—and the structural model, something that ANOVA does not permit. Second, it is statistically superior for modeling complex relationships that include multiple simultaneous moderation effects, thereby reducing measurement error (Hair et al., 2020; Sarstedt et al., 2020). Third, as a non-parametric method, it does not require multivariate normality assumptions, a condition frequently unmet when working with Likert scales and comprehension test scores (Hair et al., 2019). Furthermore,

its predictive orientation is particularly relevant given the study's objective, and its architecture allows for the integration of reflective latent constructs, manifest variables, and the nominal independent variable coded as dummies into a single model.

To incorporate PS (a categorical variable with three levels) into the structural model, dummy variable coding was implemented using the control group (1.0x) as the reference category (coded as 0, 0). Two indicator variables were created: Speed 1.5x (1 for that group, 0 for the rest) and Speed 2.0x (1 for that group, 0 for the rest). The analysis plan followed a systematic two-stage procedure (Hair et al., 2019). Stage 1 involved the evaluation of the measurement model, where indicator reliability (outer loadings > 0.708), internal consistency (Cronbach's Alpha and Composite Reliability > 0.70), convergent validity (AVE > 0.50), and discriminant validity through the HTMT criterion (Heterotrait-Monotrait Ratio < 0.85) were examined. Only once the validity of the measurement model was established did the evaluation of the structural model proceed.

Stage 2 consisted of the structural model evaluation to test the proposed hypotheses. First, the direct effects corresponding to H1 and H2 were analyzed by examining the path coefficients (β) connecting the dummy variables to the dependent variables (objective information comprehension and perceived clarity). In this context, a negative and statistically significant β coefficient compared to the reference group (1.0x) was interpreted as evidence of the detrimental effect proposed in the hypotheses. Subsequently, the moderation effects associated with H3 and H4 were evaluated using the two-stage approach recommended within the PLS-SEM framework when working with continuous moderators. This procedure involved creating interaction terms

TABLE 2 Sample distribution.

Characteristic	Category	Frequency (n)	Percentage (%)
Gender	Male	165	50.30%
	Female	162	49.39%
	Other gender	1	0.30%
Educational level	Secondary/Primary	73	22.26%
	Technical/Technological	71	21.65%
	Professional	142	43.29%
	Postgraduate	42	12.80%
Socioeconomic stratum	1	7	2.13%
	2	56	17.07%
	3	133	40.55%
	4	93	28.35%
	5	39	11.89%
Experimental group (speed)	Group 1 (Speed 1.0x)	110	33.54%
	Group 2 (Speed 1.5x)	106	32.32%
	Group 3 (Speed 2.0x)	112	34.15%
Age	Mean	39.14	
	Std. Dev.	12.21	
	Min	18	
	Max	73	

by multiplying the dummy variables by the standardized scores of technological familiarity (modeled as a construct) and age (considered a manifest variable). These interaction terms were subsequently incorporated into the structural model to estimate the moderating effects. Finally, to ensure the robustness of the findings, the statistical significance of all coefficients (p -values and confidence intervals) was determined through bootstrapping with 5,000 subsamples, providing a robust empirical estimation against potential distributional biases.

4.3 Validity and reliability

The validity and reliability of the reflective constructs (Perceived Information Clarity and Technological Familiarity) were evaluated following standard procedures (Hair et al., 2017). First, the collinearity of the indicators was examined using the Variance Inflation Factor (VIF), finding that all values were below the threshold of 5.0 (the highest was item 3 of perceived information clarity with a VIF=4.778, ruling out collinearity issues).

Regarding the reliability of the scales, as shown in Table 3, the Perceived Information construct demonstrated excellent reliability (Cronbach's Alpha=0.937; CR = 0.952) and convergent validity (AVE=0.800), with all its factor loadings above 0.80. The adjusted Technological Familiarity construct presented acceptable reliability (Cronbach's Alpha=0.691; CR = 0.826) and robust convergent validity (AVE=0.616). Although the Cronbach's Alpha is slightly below the 0.7 threshold, the Composite Reliability (CR) exceeds this minimum (Hair et al., 2017). Additionally, the items were retained since the AVE of the construct was well above 0.5 (Hair et al., 2017). Finally, discriminant validity was evaluated using the HTMT (Heterotrait-Monotrait ratio) criterion. The value obtained between the two latent constructs was 0.547, well below the strict 0.90 threshold, confirming that Perceived Information Clarity and Technological Familiarity are empirically distinct constructs.

4.4 Structural model results

Once the measurement model was validated, the structural model was evaluated using PLS-SEM to examine the proposed causal relationships. This evaluation followed

the systematic procedure recommended by Hair et al. (2017), which includes analyzing path coefficients (β), their statistical significance (p -values), the coefficient of determination (R^2), predictive relevance (Q^2), and effect size (f^2). To establish the statistical significance of the relationships, the bootstrapping technique with 5,000 subsamples was applied (Hair et al., 2017). The results of the structural model are presented below in Table 4.

The descriptive analysis of the structural model provides a detailed view of the impact of PS and the moderating variables. First, it is crucial to interpret the model's explanatory power (R^2) with caution due to the substantial differences between the two dependent variables. The model explains 30.2% ($R^2 = 0.302$) of the variance in "Perceived information clarity," indicating moderate explanatory power. In contrast, its ability to explain "objective information comprehension" is extremely weak, reaching only 2.2% ($R^2 = 0.022$). This suggests that the model explains very little variance in this outcome and that comprehension is presumably determined by other external factors not addressed in this study. Regarding the direct effect hypotheses (H1 and H2), the results are mixed. Hypothesis H1, which postulated a negative effect of speed on "objective information comprehension," was not supported by the data. Neither 1.5x speed ($\beta = 0.052$, $p = 0.419$) nor 2.0x speed ($\beta = -0.019$, $p = 0.782$) showed a statistically significant effect on objective content comprehension, and their effect sizes (f^2) were practically null.

Regarding hypothesis H2, which anticipated a negative impact on "Perceived clarity," the data offer descriptive support conditioned on speed. Increasing to 1.5x generated a marginally significant decrease ($\beta = -0.086$, $p = 0.055$); however, its effect size ($f^2 = 0.008$) falls below the minimum threshold of 0.02, which should be interpreted as a lack of practical significance. On the other hand, the effect of 2.0x speed was considerably more pronounced, showing a statistically significant negative impact ($\beta = -0.311$, $p = 0.000$) and a medium or practically relevant effect size ($f^2 = 0.104$). Analyzing control and moderating variables, "Technological Familiarity" had no direct effect on "objective information comprehension" ($\beta = 0.092$, $p = 0.166$), but emerged as the strongest positive predictor of "Perceived clarity" ($\beta = 0.471$, $p = 0.000$) with a considerable effect size ($f^2 = 0.309$). "Age" presented a similar pattern: no effect on "objective information comprehension" ($\beta = -0.055$, $p = 0.341$), but a positive direct effect on "Perceived clarity"

TABLE 3 Validity and reliability results of reflective constructs.

Construct	Indicator	Loading	Alpha	CR	AVE
Perceived information clarity (PC)	Cp1	0.919	0.937	0.952	0.800
	Cp2	0.904			
	Cp3	0.932			
	Cp4	0.906			
	Cp5	0.806			
Technological familiarity (FT)	Ft2	0.815	0.691	0.826	0.616
	Ft3	0.872			
	Ft4	0.649			

For reflective indicators (R), it is expected that Cronbach's Alpha > 0.7, Composite Reliability (CR) > 0.7, factor loading > 0.5, and Average Variance Extracted (AVE) > 0.5.

TABLE 4 Structural model results.

Path relationship	Information comprehension (R ² = 0.022)	Perceived information clarity (R ² = 0.302)
Direct effects (H1, H2)	β (t-value, <i>p</i> -value) [<i>f</i> ²]	β (t-value, <i>p</i> -value) [<i>f</i> ²]
1.5x (vs. 1.0x)	0.052 (t = 0.809, <i>p</i> = 0.419) [0.002]	−0.086 (t = 1.925, <i>p</i> = 0.055) [0.008]
2.0x (vs. 1.0x)	−0.019 (t = 0.277, <i>p</i> = 0.782) [0.000]	−0.311 (t = 6.164, <i>p</i> = 0.000) [0.104]
Controls (direct effects)		
Age	−0.055 (t = 0.954, <i>p</i> = 0.341) [0.003]	0.117 (t = 2.365, <i>p</i> = 0.018) [0.019]
Technological familiarity	0.092 (t = 1.389, <i>p</i> = 0.166) [0.009]	0.471 (t = 8.821, <i>p</i> = 0.000) [0.309]
Technological familiarity moderation (H3)		
TF × 1.5x	−0.073 (t = 0.993, <i>p</i> = 0.321) [0.004]	0.001 (t = 0.018, <i>p</i> = 0.986) [0.000]
TF × 2.0x	−0.006 (t = 0.075, <i>p</i> = 0.940) [0.000]	−0.029 (t = 0.399, <i>p</i> = 0.690) [0.001]
Age moderation (H4)		
Age × 1.5x	0.013 (t = 0.200, <i>p</i> = 0.842) [0.000]	0.111 (t = 2.752, <i>p</i> = 0.006) [0.013]
Age × 2.0x	0.044 (t = 0.656, <i>p</i> = 0.512) [0.001]	0.059 (t = 0.976, <i>p</i> = 0.330) [0.004]

Exact *p*-values are reported. Thresholds for effect sizes [*f*² are: small (*f*² ≥ 0.02), medium (*f*² ≥ 0.15), and large (*f*² ≥ 0.35)]. Values below 0.02 indicate a lack of practical significance.

($\beta = 0.117$, $p = 0.018$), albeit with low practical significance ($f^2 = 0.019$). Finally, moderation hypotheses (H3 and H4) were evaluated. Hypothesis H3 was not supported, as “Technological Familiarity” did not significantly interact with speeds for either comprehension or clarity ($p > 0.300$ in all cases). Hypothesis H4, regarding “Age,” received partial statistical support. A statistically significant interaction ($\beta = 0.111$, $p = 0.006$) was observed between Age and the 1.5x speed condition on “Perceived clarity,” descriptively indicating that age alters how clarity is perceived at this speed. Nevertheless, the size of this interaction effect is very small ($f^2 = 0.013$), placing it below the threshold of practical relevance. Below, Figure 2 illustrates the estimated simple slopes for each age group across the three speed conditions.

As illustrated in Figure 2, the three slopes descend as speed increases, which is consistent with the general negative effect on perceived clarity. However, the magnitude of the decrease differs by age group. Younger participants (≈ 27 years) exhibit the most pronounced drop, with perceived clarity values of 4.10 at 1.0x, 3.91 at 1.5x, and 3.75 at 2.0x. In contrast, older participants (≈ 51 years) show a notably different pattern: their perceived clarity remains stable between 1.0x ($M = 4.35$) and 1.5x ($M = 4.38$), with a slight drop only at 2.0x ($M = 4.10$). This pattern suggests that age might act as a factor that buffers the negative effect of moderately increased speeds. Nevertheless,

given that the effect size is small ($f^2 = 0.013$) and the Ages $\times$ 2.0x interaction was non-significant ($p = 0.330$), these results must be interpreted with caution.

4.5 Evaluation of the structural model

To assess the predictive relevance and overall fit of the model, the Stone-Geisser Q^2 , Goodness of Fit (GoF), and Standardized Root Mean Square Residual (SRMR) indices were analyzed, with results presented in Table 5.

Regarding predictive relevance (Q^2), the blindfolding procedure proposed by Chin et al. (2008) was applied. The Q^2 value for the “Perceived Information Clarity” variable was positive and substantial ($Q^2 = 0.235$), indicating adequate predictive capacity of the model for this variable. In contrast, the value obtained for the “objective information comprehension” variable was negative ($Q^2 = -0.041$), indicating a lack of predictive relevance, consistent with the low coefficient of determination (R^2) associated with this variable.

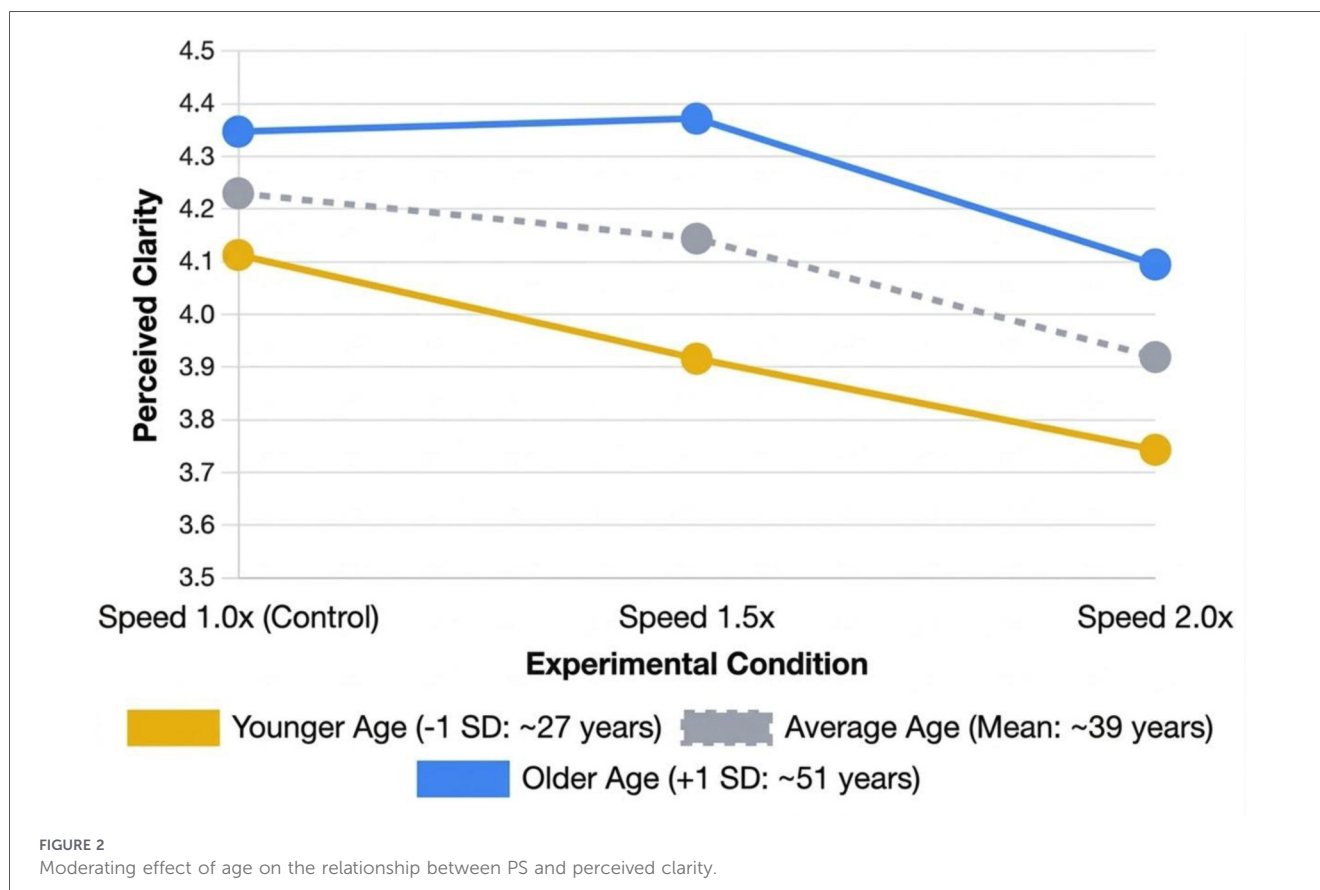
Regarding the overall fit of the model (GoF), it was calculated following the proposal by Wetzels et al. (2009), obtaining a value of 0.338, which suggests a medium-to-high fit, above the 0.25 threshold considered representative of a medium level of fit.

Finally, regarding the model fit (SRMR), the obtained value was 0.067. Since this value is below the 0.08 threshold established by Hu and Bentler (1999), it is concluded that the model presents a good overall fit to the data.

Regarding predictive relevance (Q^2), the blindfolding procedure proposed by Chin et al. (2008) was applied. The (Q^2), value for “Perceived information clarity” was positive and substantial ($Q^2 = 0.235$), demonstrating adequate predictive capacity of the model for this construct. In contrast, the value obtained for “objective information comprehension” was negative ($Q^2 = -0.041$), indicating a total absence of predictive relevance; a result that is directly consistent with the low coefficient of determination (R^2) associated with this variable. Regarding the global Goodness of Fit (GoF), it was calculated following the approach by Wetzels et al. (2009), yielding a value of 0.338. This suggests a medium-to-high level of fit, as it exceeds the 0.25 threshold. Finally, concerning the standardized root mean square residual (SRMR), the value obtained was 0.067. Since this indicator is below the conservative threshold of 0.08 established by Hu and Bentler (1999), it is concluded that, in global terms, the model presents a good fit to the empirical data obtained.

5 Discussion

The results of this study offer a nuanced and, in certain respects, counterintuitive view of the PS phenomenon, situating the findings at the intersection of cognition, perception, and individual differences. The most significant contribution of this research lies in the observed dissociation between objective information comprehension and perceived information clarity—a dualism that, far from being contradictory, can be interpreted as a manifestation of the differentiated levels of processing described by Cognitive Load Theory (CLT; Sweller, 1988): while long-term memory encoding can be preserved under moderate load, the



processing fluency experienced subjectively as clarity deteriorates before objective retention collapses. The central finding is that H1 was not supported and H2 received partial statistical support conditioned on speed. The results indicate that PS, even at 2.0x, did not have a statistically significant impact on objective content comprehension. This result is consistent with a strand of literature documenting certain cognitive tolerance to time compression (Chen et al., 2024; Kiyak et al., 2023), although it contradicts studies that reported performance impairments (Liu and Jia, 2025; Tharumalingam et al., 2025). A possible explanation, coherent with CLT, is that the speeds employed in this study did not exceed the individual working memory overload threshold necessary to disrupt encoding; instead, they were sufficient to increase extraneous load and degrade perceptual fluency. Nevertheless, it is important to note that comprehension was measured immediately through a brief test, which limits the generalization of this finding: it is possible that the negative

effects of PS emerge in tasks of greater complexity, in delayed evaluations, or in content with higher conceptual density.

The near-zero explanatory power of the model regarding comprehension ($R^2 = 0.022$) should be treated as a central explanatory limitation of this study. The proposed model explains very little variance in this outcome, suggesting that the included variables (speed, age, and technological familiarity) are insufficient to predict performance on the knowledge test. This is theoretically coherent with Kintsch's (1988) postulates on limited processing: objective information comprehension in tasks of moderate duration might be primarily determined by dispositional variables not measured here, such as individual working memory capacity, rather than by the experimental manipulation of speed. In marked contrast, H2 received statistical support conditioned on speed, revealing a dose-response relationship between acceleration and perceptual deterioration. This pattern is consistent with the threshold logic implicit in CLT (Sweller, 1988): at 1.5x, the additional extraneous load was marginally perceptible but without practical relevance, whereas at 2.0x, the temporal compression imposed a sustained demand on attentional resources that participants experienced as arduous, non-fluent, and difficult-to-follow processing. It is notable that this perceptual deterioration occurs in the absence of equivalent damage to objective information comprehension. Following Keehr and Reardon (2025), this gap could have relevant metacognitive implications: users who experience low clarity might underestimate their own learning, generating an "illusion of non-learning" with potential consequences for self-efficacy and motivation. However, since this study did not directly measure metacognitive beliefs or perceived self-efficacy after exposure, this

TABLE 5 Evaluation of the structural model.

Metric	Construct	Value	Criterion
Q ² (CVR)	Perceived information clarity	0.235	> 0 (Substantial)
	Information comprehension	-0.041	< 0 (Null)
GoF	(Global fit)	0.338	> 0.25 (Medium)/ ~0.36 (High)
SRMR	(Standardized root mean square residual)	0.067	< 0.08 (Good Fit)

Q² (Chin et al., 2008), GoF (Wetzels et al., 2009), SRMR (Hu and Bentler, 1999).

interpretation should be considered speculative and constitutes a priority direction for future research.

Regarding the moderation hypotheses, H3 was not supported: technological familiarity (TF) did not moderate the relationship between PS and either of the dependent variables, despite emerging as the strongest direct predictor of perceived clarity. This result nuances the postulates of Computer Self-Efficacy Theory (Compeau and Higgins, 1995): individuals with high TF seem to have developed automated schemas to process information in digital environments, which translates into higher perceived clarity in baseline conditions. However, this proficiency does not confer differential protection against speed increases. In other words, TF raises the perceptual starting point but does not buffer the slope of deterioration when speed increases, suggesting that the impact of PS on clarity operates relatively independently of the user's prior digital competence.

H4 obtained partial and limited statistical support. The only significant interaction found (between Age and the 1.5x condition on perceived clarity) indicates that the negative effect of moderate speed was less pronounced in older participants, which is counterintuitive relative to Processing-Speed Theory (Salthouse, 1996), which predicts greater cognitive vulnerability in older adults facing accelerated processing demands, and relative to previous findings that reported greater impact in this group (Murphy et al., 2023). This finding allows for at least two alternative readings that must be considered with caution. On one hand, it could reflect that older participants, who reported slightly higher baseline clarity, possess more flexible processing expectations or lower metacognitive sensitivity to moderate speed increments (Tharumalingam and Risko, 2025). On the other hand, it could be due to uncontrolled differences in prior video content consumption patterns or familiarity with the experimental task. In any case, given that the effect size is small and the interaction did not replicate in the higher acceleration condition, these conclusions should be treated as exploratory and require verification in larger and more diverse samples.

6 Conclusions

This study provides empirical evidence of the dissociation between two dimensions of learning frequently treated as equivalent: objective information comprehension and perceived information clarity. Under the specific conditions of this study (a technical lesson of moderate duration evaluated immediately), increasing PS up to 2.0x did not significantly impair objective content retention, although it did produce a progressive and statistically robust deterioration in the perception of clarity, especially at 2.0x. This pattern suggests that PS operates as an asymmetric exchange: it preserves the learning outcome but increases the cost of its subjective experience. Regarding individual differences, neither technological familiarity nor age moderated the impact of PS on comprehension. Technological familiarity predicted higher perceived clarity in baseline conditions but did not buffer its deterioration in the face of acceleration. The only significant interaction found (age $\times$ 1.5x on perceived clarity) was of small magnitude and should be treated as an exploratory finding that requires replication.

6.1 Implications practice

For students using PS as a time management strategy, the findings of this study suggest that, at least for technical content of moderate duration, acceleration does not necessarily compromise objective retention. However, the feeling of confusion accompanying high speeds is real and could lead to underestimating one's own learning, making it advisable for students to develop active verification strategies for their comprehension. For educators and instructional designers, these results point to the importance of structuring content with clear visual supports and explicit segmentation, especially if it is anticipated that the material will be consumed at accelerated speeds.

6.2 Limitations and future directions

This study presents limitations that narrow the scope of its conclusions. First, the exposure was brief (17 min) and confined to a specific technical topic; therefore, the findings cannot be extrapolated to materials of longer duration, greater conceptual abstraction, or different disciplines. Second, comprehension was measured immediately through a brief test, which precludes conclusions regarding delayed retention; it is possible that the effects of PS on long-term learning differ from those observed here. Third, the model explained very little variance in objective information comprehension ($R^2 = 0.022$), indicating that unmeasured variables, such as individual working memory capacity, likely carry greater explanatory weight than the manipulated variables. Finally, perceived clarity was measured through self-report, which introduces the limitations inherent to subjective methods, including potential social desirability or response biases. Future research should examine whether there is a duration or complexity threshold at which comprehension is finally compromised, evaluate delayed retention after periods of days or weeks, control for individual working memory capacity as a covariate, and complement self-report measures with behavioral or physiological indicators of cognitive effort.

Data availability statement

The datasets presented in this study can be found in online repositories. The names of the repository/repository and accession number(s) can be found below: The data presented in this study are openly available in ZENODO at <https://doi.org/10.5281/zenodo.17783418>, reference number 17783418.

Ethics statement

The studies involving humans were approved by Ethics Committee for Research of the Faculty of Administrative Sciences (CEICA), Universidad del Valle, Cali, Colombia. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

AR: Project administration, Resources, Supervision, Investigation, Writing – review & editing, Conceptualization, Funding acquisition. EA: Validation, Writing – review & editing, Supervision, Investigation, Visualization, Conceptualization, Resources. CO-A: Writing – original draft, Formal analysis, Validation, Methodology, Software, Investigation.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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